MATHEMATICS / INDEPENDENT STUDY

MM7030

Category Theory

Complete solutions

Original teaching material aligned with Stockholm University course descriptions.
Independently authored; not an official university publication.

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How to use this volume E problems practice definitions and standard arguments; C problems combine ideas; A problems require synthesis or generalization. These are study difficulty labels, not official SU grading criteria. Time estimates are guides, not deadlines. Write a complete attempt before opening hints. In the hints volume, read Hint 1 first and return to the problem before consulting Hint 2. Solutions give an argument and criteria for self-assessment. Expanded solutions also explain how to find the approach, a tempting dead end, and the role of the hypotheses.

Each unit has twelve core problems (four E, five C, three A) and an optional A-level transfer challenge numbered 13. The challenge asks you to choose a method in a new setting; budget its hour separately. Problem 12 and the challenge have detailed solutions with independently written decisive ideas. Mock estimates total 170 minutes of problem work plus ten minutes for review.

1 Unit 1 · Categories and duality

MM7030-U1.01 · E · 15 min
Learning objective: U1.1 Category axioms

Verify that nonnegative integers with an arrow $m \to n$ exactly when $m \leq n$ form a category.

Reflexivity supplies identities; transitivity supplies a composite whenever arrows are composable. Both possible associative composites have the same source and target and hence are the unique arrow of that type. Identity laws hold for the same reason.

Check your argument. Check identities, existence of composition, and associativity.

MM7030-U1.02 · E · 15 min
Learning objective: U1.1 One-object categories

Describe the one-object category associated with the additive monoid $\mathbb{N}$. What arrows are invertible?

There is one object, arrows are nonnegative integers, composition is addition, and the identity arrow is zero. Only zero is invertible, since $n + m = 0$ with both nonnegative forces both zero.

Check your argument. Distinguish the object from its many endomorphisms.

MM7030-U1.03 · E · 15 min
Learning objective: U1.2 Opposites

Describe the opposite of a poset category.

The opposite has the same objects with the reversed order. Identities are unchanged, and transitivity of the reversed order supplies composition.

Check your argument. State the new arrow condition explicitly.

MM7030-U1.04 · E · 15 min
Learning objective: U1.1 Isomorphisms

Prove an isomorphism in the category of groups is exactly a bijective group homomorphism.

An inverse morphism makes the underlying function bijective. Conversely let $f : G \to H$ be a bijective homomorphism. For $x = f(a), y = f(b)$, $f^{-1}(xy) = f^{-1}(f(ab)) = ab = f^{-1}(x)f^{-1}(y)$. Thus its inverse function is a homomorphism.

Check your argument. Prove the reverse direction rather than treating inverse preservation as automatic.

MM7030-U1.05 · C · 30 min
Learning objective: U1.2 Monomorphisms

Prove the composite of two monomorphisms is monic.

If n, m are monic and $(nm)u = (nm)v$, monicity of n gives $mu = mv$, then monicity of m gives $u = v$. This is the definition of monicity of the composite.

Check your argument. State the sources and targets needed for all composites.

MM7030-U1.06 · C · 30 min
Learning objective: U1.2 Cancellation

If gf is monic, prove f is monic. Must g be monic?

If $fu = fv$, then $gfu = gfv$, so monicity of gf gives $u = v$. The second claim fails: let f include a singleton into a two-element set and let g be the constant map to a singleton. The composite is a bijection on a singleton, hence monic, but g is not injective.

Check your argument. Prove the implication and supply a fully typed counterexample.

MM7030-U1.07 · C · 30 min
Learning objective: U1.2 Split arrows

Show a morphism with a left inverse is monic. Give a monomorphism of sets with no left inverse.

If $rf = 1$, then $fu = fv$ implies $u = rfu = rfv = v$. The unique map $\varnothing \to 1$ is injective and therefore monic, but it has no left inverse because no map $1 \to \varnothing$ exists.

Check your argument. A left inverse must be an actual arrow with the required source.

MM7030-U1.08 · C · 30 min
Learning objective: U1.2 Epimorphisms

Prove a function missing an element of its codomain is not epic in sets.

If $f : X \to Y$ misses y_0, let $u : Y \to \{0, 1\}$ be constantly zero and let v be one at y_0 and zero elsewhere. Then $u \neq v$ but $uf = vf$, because f never reaches y_0. Hence f fails the epi cancellation property.

Check your argument. Specify the test maps and why their composites agree.

MM7030-U1.09 · C · 30 min
Learning objective: U1.1 Groupoids

Show the isomorphisms in any category, retaining all objects, form a subcategory in which every arrow is invertible.

Identities are isomorphisms. For isomorphisms $f : A \to B$ and $g : B \to C$, the composite gf has proposed inverse $f^{-1}g^{-1}$. It satisfies both inverse equations by associativity, so composites of isomorphisms remain isomorphisms. The inverses themselves are isomorphisms, hence lie in this subcategory. It is a groupoid.

Check your argument. Verify the composition order.

MM7030-U1.10 · A · 50 min
Learning objective: U1.2 Balancedness

Prove a morphism that is both monic and split epic is an isomorphism.

Let $f : A \to B$ and choose its right inverse $s : B \to A$. Since $fs = 1_B$, we have $f(sf) = (fs)f = f = f1_A$. Monicity of f gives $sf = 1_A$. Thus s is both a right and left inverse, and f is an isomorphism.

Check your argument. Do not assume an arbitrary epi has a section.

MM7030-U1.11 · A · 50 min
Learning objective: U1.2 Rings versus sets

Prove the inclusion $\mathbb{Z} \to \mathbb{Z}\left[\frac{1}{2}\right]$ is epic among commutative unital rings but is not an isomorphism.

If two homomorphisms out of $\mathbb{Z}\left[\frac{1}{2}\right]$ agree on $\mathbb{Z}$, their values on $\frac{1}{2}$ are both inverses of the common image of two, hence agree. Every element is $\frac{a}{2^n}$, so all their values agree. This proves the inclusion epic. It is not onto because $\frac{1}{2}$ is not an integer, and a ring isomorphism must be a bijection.

Check your argument. Quantify over every possible target ring and compare all fractions.

MM7030-U1.12 · A · 50 min
Learning objective: U1.1 Categories from paths

Construct the free category on a directed graph and prove any assignment of its vertices and edges to objects and arrows of a category extends uniquely to a functor.

Decisive idea. Keep paths formal until evaluating them in the target; the functor axioms then force every value.

Finding the approach. The graph has generators but no composites yet. Build the missing arrows formally, so that no accidental equation is imposed beyond the category axioms.

Argument. Objects are vertices; arrows are finite composable paths; empty paths are identities; composition concatenates paths. Concatenation is associative. Given a graph assignment preserving endpoints, send each path to the composite of its edge images and each empty path to an identity. Associativity and identity laws in the target prove this is a functor. Every path is a composite of edges, so any extension must have these values, proving uniqueness.

A path has specified initial and terminal vertices; only matching endpoints may be concatenated. Parenthesizing concatenations does not change the edge sequence. For an image of a length-three path, the two possible target composites agree by associativity, and induction handles every finite length. Empty paths must map to identities, so the construction also covers isolated vertices. Uniqueness is equality on all paths, not just on the graph's edges.

A tempting dead end. Do not identify different paths merely because they have the same endpoints. That produces a thin category and destroys the freedom to map parallel paths to different arrows.

Hypotheses. The graph assignment must preserve endpoints. The target's associativity and identity laws are what make evaluation of formal paths a functor.

Check your argument. Include empty paths and explain why no additional path relations are imposed.

MM7030-U1.13 · A · 60 min · Optional challenge

Learning objective: U1 Transfer: Words as arrows

Form a category with one object whose arrows are finite words in the two symbols a, b, including the empty word; composition is concatenation. Verify the category laws, determine which arrows are monomorphisms and epimorphisms, and determine which are isomorphisms.

Decisive idea. Use cancellation to analyze mono and epi, then a different invariant—length—to analyze inverses.

Finding the approach. In a one-object category, cancellation definitions are equations between words. Length gives an independent obstruction to invertibility.

Argument. Associativity holds because concatenating three words produces the same ordered sequence under either parenthesization. The empty word is a two-sided identity. If $wu = wv$, delete the identical initial segment w to obtain $u = v$; if $uw = vw$, compare lengths first and then delete the identical final segment to obtain $u = v$. Thus every arrow is both monic and epic, regardless of the chosen convention for which factor is applied first. If w has an inverse v, their concatenation is empty and $|w| + |v| = 0$. Both lengths are nonnegative, so w is empty. Conversely the empty word is its own inverse. In particular a nonempty word is monic and epic without being an isomorphism.

A tempting dead end. The implication 'monic and epic means invertible' is true in sets but not in every category. This category provides the missing counterexample.

Hypotheses. Arrows are words themselves, not words modulo additional relations. Quotienting by relations can destroy cancellation or introduce inverses.

Check your argument. Check both cancellation directions and distinguish them from existence of an inverse.

2 Unit 2 · Functors and equivalences

MM7030-U2.01 · E · 15 min
Learning objective: U2.1 Functor laws
Show a monotone map between posets gives a functor between their categories.
If $a \leq b$, monotonicity gives $F(a) \leq F(b)$, so the unique arrow maps to the unique arrow. Identities and composites are preserved because the relevant hom sets contain at most one arrow.
Check your argument. Explain why monotonicity is exactly the condition needed.

MM7030-U2.02 · E · 15 min
Learning objective: U2.1 Forgetful functors
Define the forgetful functor from vector spaces over K to sets on objects and arrows.
It sends a vector space to its underlying set and a linear map to its underlying function. Identity linear maps are identity functions and linear-map composition is function composition, proving the functor laws.
Check your argument. Describe both assignments.

MM7030-U2.03 · E · 15 min
Learning objective: U2.1 Hom functors
For $f : X \to Y$, write the map $\mathcal{C}(A, f)$.
It is the function $\mathcal{C}(A, X) \to \mathcal{C}(A, Y)$, $h \mapsto fh$. The types ensure the composite exists.
Check your argument. Give domain, codomain, and formula.

MM7030-U2.04 · E · 15 min
Learning objective: U2.2 Faithfulness
Explain why the forgetful functor from groups to sets is faithful.
If two group homomorphisms have equal underlying functions, they take the same value at every group element and are the same homomorphism. Thus each induced hom-set map is injective.
Check your argument. Use the hom-set definition of faithful.

MM7030-U2.05 · C · 30 min
Learning objective: U2.1 Contravariance
Verify inverse image of subsets is a contravariant functor on sets.
For $f : X \to Y$, $g : Y \to Z$ and $S \subseteq Z$, $x \in (gf)^{-1}S$ exactly when $g(f(x)) \in S$, equivalently $x \in f^{-1}(g^{-1}S)$. Thus composition reverses. Identity inverse image is the identity on the power set.
Check your argument. State the reversed direction of the arrow map.

MM7030-U2.06 · C · 30 min
Learning objective: U2.2 Fullness
Show the forgetful functor from groups to sets is not full.
The constant function $C_2 \to C_2$ with value its nonidentity element is a set function but cannot be a group homomorphism, since homomorphisms send identity to identity. Therefore the hom-set map is not surjective.
Check your argument. Use a map between underlying sets of actual objects in the source category.

MM7030-U2.07 · C · 30 min
Learning objective: U2.2 Reflection of isomorphisms
Prove a full and faithful functor reflects isomorphisms.
If Ff has inverse u, fullness gives g with $Fg = u$. Then $F(gf) = FgFf = 1 = F(1)$ and similarly $F(fg) = F(1)$. Faithfulness gives $gf = 1$ and $fg = 1$, so f is an isomorphism.
Check your argument. Specify which property supplies the inverse candidate and which verifies its equations.

MM7030-U2.08 · C · 30 min
Learning objective: U2.1 One-object functors
Show functors between one-object categories of monoids are precisely monoid homomorphisms.
The unique object must map to the unique object. The arrow map is a function between monoids. The two functor laws require it to preserve the unit and multiplication, exactly the monoid-homomorphism

axioms. Conversely those axioms define a functor.
Check your argument. State both directions of the identification.

MM7030-U2.09 · C · 30 min
Learning objective: U2.2 Matrix category
Show the category of matrices over K, with objects nonnegative integers, is equivalent to finite-dimensional vector spaces.
The matrix-to-linear-map functor preserves multiplication and identities. The uniqueness and existence of coordinate matrices make it full and faithful. Every finite-dimensional vector space has a finite basis and is isomorphic to some K^n, giving essential surjectivity. The equivalence criterion applies.
Check your argument. Include dimension zero and distinguish essential from literal surjectivity.

MM7030-U2.10 · A · 50 min
Learning objective: U2.2 Faithful reflection
Prove a faithful functor reflects monomorphisms and epimorphisms, without assuming fullness.
If Ff is monic and $fu = fv$, then $FfFu = FfFv$, so $Fu = Fv$ and faithfulness gives $u = v$. Thus f is monic. Dually, if Ff is epic and $uf = vf$, cancel Ff on the right and use faithfulness to conclude $u = v$.
Check your argument. Do not confuse reflection with preservation; fullness is unnecessary here.

MM7030-U2.11 · A · 50 min
Learning objective: U2.2 Equivalence of posets
Show equivalent poset categories are isomorphic as posets.
If F, G witness an equivalence, the component isomorphisms $GF(p) \cong p$ and $FG(q) \cong q$ force literal equality by antisymmetry. Thus the monotone maps are inverse bijections. Each direction is monotone, so they are an order isomorphism.
Check your argument. Use antisymmetry; the statement need not hold for preorders.

MM7030-U2.12 · A · 50 min
Learning objective: U2.2 Failure of preservation
Construct a faithful functor which sends a monomorphism and epimorphism to a function which is neither. Use the two-object poset $0 < 1$ as source, and explain why this does not contradict U2.10.
Decisive idea. Identify the missing target test arrows to understand exactly why reflection does not imply preservation.
Finding the approach. Faithfulness only compares arrows already present in each source hom set. A thin source makes that requirement easy to satisfy even when its target has many more test arrows.
Argument. Map both objects to the set $\{a, b\}$, identities to identities, and $0 \to 1$ to the constant function with value a. These assignments form a functor because every nontrivial composition involves an identity. It is faithful: each source hom set is empty or a singleton, so its map into the corresponding function set is injective. The nonidentity source arrow is both monic and epic, but its image is neither injective nor surjective, hence neither monic nor epic in sets. Faithfulness reflects cancellation properties from the target; it does not preserve them from the source.
To see failure of monicity in sets, choose the two distinct functions from a singleton selecting a and b; composing either with the constant map gives the same function. To see failure of epicity, choose two functions from the two-point codomain to itself that agree at a but differ at b. Their composites with the constant map agree. These witnesses are absent from the relevant source hom sets, so faithfulness cannot rule them out.
A tempting dead end. A faithful functor need not be full, injective on objects, or preserve every categorical property. The direction of a reflection argument cannot be reversed.
Hypotheses. The source is a poset viewed as a category, so every parallel pair there is equal. Cancellation in the target must be tested against all target arrows, not only image arrows.
Check your argument. Verify functoriality and faithfulness, and distinguish preservation from reflection.

MM7030-U2.13 · A · 60 min · Optional challenge

Learning objective: U2 Transfer: Discarding redundant points

Let objects be pairs (X, e) with X a finite set and $e : X \to X$ satisfying $e^2 = e$. An arrow $f : (X, e) \to (Y, d)$ satisfies $fe = f = df$, and the identity of (X, e) is e. Verify this is a category. Show that sending (X, e) to its image $e(X)$ and arrows to restrictions gives an equivalence with finite sets.

Decisive idea. Restriction and extension are inverse descriptions of arrows, while every target object already occurs as an image.

Finding the approach. The unusual identity tells us that points outside the image carry no independent information. Restriction should remove exactly that redundancy, and the displayed extension tests whether any information is lost.

Argument. Compositions satisfy the same equations: if $fe = f = df$ and $gd = g = cg$, then $(gf)e = gf$ and $c(gf) = gf$. Composition is associative as composition of functions, and e, d act as the required identities on f. Every f maps into $d(Y)$ and is determined by its restriction to $e(X)$ because $f(x) = f(e(x))$. Conversely any function $h : e(X) \to d(Y)$ extends to $f(x) = h(e(x))$; idempotence gives both arrow equations. Restriction is therefore a bijection on each hom set, hence full and faithful. Every finite set Z is the image of (Z, id_Z), so the functor is essentially surjective and an equivalence. The inclusion and retraction between (X, e) and $(e(X), \mathrm{id})$ have composites e and the identity, exactly the identities in this category.

A tempting dead end. The identity on the underlying set X is generally not an arrow of this category. Using it in place of e breaks the verification.

Hypotheses. Idempotence is essential both for the category's identities and for the extension formula. Finiteness keeps the target within the chosen category but is not used in the hom-set bijection.

Check your argument. Verify the category first, then prove fullness, faithfulness, and essential surjectivity explicitly.

3 Unit 3 · Natural transformations

MM7030-U3.01 · E · 15 min
Learning objective: U3.1 Naturality
Write the naturality equation for $\eta : F \Rightarrow G$ at $f : A \to B$.
The equation is $G(f)\eta_A = \eta_B F(f)$. The left path goes first across at A and then down by Gf; the right path first uses Ff and then the component at B.
Check your argument. Check the domain and codomain of each factor.

MM7030-U3.02 · E · 15 min
Learning objective: U3.2 Identities
Show the identity maps 1_{FA} form a natural transformation $F \Rightarrow F$.
$Ff1_{FA} = Ff = 1_{FB}Ff$ by identity laws. Thus the component family is natural.
Check your argument. Use identity arrows at the correct objects.

MM7030-U3.03 · E · 15 min
Learning objective: U3.1 Natural projections
Show $\pi_1 : X \times X \to X$ is natural in the set X.
Both $f\pi_1$ and $\pi_1(f \times f)$ send (x, y) to $f(x)$. Hence the square commutes for every function f, proving naturality.
Check your argument. Quantify over all functions.

MM7030-U3.04 · E · 15 min
Learning objective: U3.2 Components
Describe a natural transformation between two constant functors with values A, B when the source is the terminal one-object category.
It is any arrow $A \to B$. The sole naturality equation is $1_B\eta = \eta 1_A$, automatically true.
Check your argument. Explain why no additional compatibility condition occurs.

MM7030-U3.05 · C · 30 min
Learning objective: U3.1 Diagonal
Show the diagonal $\Delta_X : X \to X \times X$ is natural and each component is injective.
For any $f : X \to Y$, $(f \times f)\Delta_{X(x)} = (f(x), f(x)) = \Delta_Y f(x)$, proving naturality. If $\Delta_{X(x)} = \Delta_{X(y)}$, equality of first coordinates gives $x = y$, so each component is injective.
Check your argument. Prove naturality separately from injectivity.

MM7030-U3.06 · C · 30 min
Learning objective: U3.2 Vertical composition
Prove vertical composition of natural transformations is associative.
For composable η, θ, κ, the components of $(\kappa\theta)\eta$ and $\kappa(\theta\eta)$ are $(\kappa_A\theta_A)\eta_A$ and $\kappa_{A(\theta_A\eta_A)}$, equal by associativity. Transformations are equal when all components agree, so composition is associative.
Check your argument. Include the componentwise equality criterion.

MM7030-U3.07 · C · 30 min
Learning objective: U3.1 Inverse naturality
If $\eta : F \Rightarrow G$ is natural and each component is invertible, prove the inverse family is natural.
Compose on the left with η_B^{-1} and on the right with η_A^{-1} to obtain $\eta_B^{-1}Gf = Ff\eta_A^{-1}$. This is the naturality equation for a transformation $G \Rightarrow F$. Componentwise inverse equations make it the inverse transformation.
Check your argument. Keep the indices consistent.

MM7030-U3.08 · C · 30 min
Learning objective: U3.2 Group actions
For two actions of a group G on sets X, Y, show natural transformations of the corresponding functors are exactly equivariant functions.
Because the group category has one object, the component family is one function T. Naturality at each arrow g says $T(g \cdot x) = g \cdot T(x)$ for every x. These are precisely the equations defining an equivariant

function. Conversely an equivariant function satisfies every naturality square.
Check your argument. State both directions.

MM7030-U3.09 · C · 30 min
Learning objective: U3.1 Double dual
Verify naturality of the canonical map $V \to V^{**}$ by evaluating on a linear functional.
Let $\eta_{V(v)}(\varphi) = \varphi(v)$. Then $f^{**}\left(\eta_{V(v)}\right)(\psi) = \eta_{V(v)}(\psi f) = \psi(f(v)) = \eta_{W(f(v))}(\psi)$. Equality for every ψ gives $f^{**}\eta_V = \eta_W f$.
Check your argument. State the types of v, ψ, and the dual maps.

MM7030-U3.10 · A · 50 min
Learning objective: U3.1 Natural endomorphisms of sets
Classify all natural transformations from the identity functor on sets to itself without using Yoneda.
Let η be natural. For $x \in X$, let $i_x : 1 \to X$ select x. The only function $1 \to 1$ is the identity, so naturality gives $\eta_{X(x)} = \eta_X i_{x(*)} = i_x \eta_1(*) = x$. Thus every component is the identity, including the unique map on the empty set. The identity family is natural, so it is the only one.
Check your argument. Handle arbitrary elements via naturality, not a choice of ordering.

MM7030-U3.11 · A · 50 min
Learning objective: U3.2 Center of a group
Identify the natural endomorphisms of the identity functor of a group regarded as a one-object category.
A transformation is an element $z \in G$ with $gz = zg$ for every g. Thus its elements are exactly $Z(G)$. Vertical composition is multiplication of these components, and every component is invertible, so these transformations form the center as a group.
Check your argument. Describe both the set of transformations and its composition.

MM7030-U3.12 · A · 50 min
Learning objective: U3.1 Scalar natural operations (proof reconstruction)
Show every natural endomorphism of the identity functor on K-vector spaces is multiplication by a single scalar.
Decisive idea. The component at the free one-generator object determines every other component through maps carrying its generator to an arbitrary vector.
Finding the approach. Use a one-dimensional test object. Every vector is the image of its distinguished element under a linear map, so naturality propagates one scalar to every space.
Argument. Let $c = \eta_{K(1)}$. For $v \in V$, the map $i_v : K \to V$, $a \mapsto av$, is linear. Naturality gives $\eta_{V(v)} = \eta_V i_{v(1)} = i_v \eta_{K(1)} = cv$. Thus all components use the same scalar. Conversely scalar multiplication commutes with every linear map, so every $c \in K$ yields such a transformation.
Naturality of i_v is the equation $\eta_V \circ i_v = i_v \circ \eta_K$. Evaluating at one is legitimate because both sides are functions from K to V. The scalar c is chosen once from the component at K, not separately for each V. The zero vector space causes no exception: it has only one endomorphism and scalar multiplication is that map. Conversely, for any linear f, $f(cv) = cf(v)$ verifies the full naturality square.
A tempting dead end. Checking that each component is some scalar multiple of the identity does not establish that the scalar is the same for all spaces. Naturality is the compatibility that forces uniformity.
Hypotheses. Components and test maps are linear, and the category contains K and all maps i_v. Restricting the available test maps could allow more transformations.
Check your argument. Prove the scalar is independent of both the vector and the space.

MM7030-U3.13 · A · 60 min · Optional challenge
Learning objective: U3 Transfer: Uniform operations on subsets
Let P send a set to its power set and a function to direct image. Classify all natural transformations $P \Rightarrow P$ on the category of sets.
Decisive idea. Use symmetry to limit local possibilities, then noninvertible maps to force one global choice.
Finding the approach. Naturality under permutations forces any output attached to the whole set to treat all points alike. A map to the singleton synchronizes the choice across different sets.
Argument. Let η be natural. The unique map from the empty set to X implies $\eta_{X(\varnothing)} = \varnothing$. Write $S_A = \eta_{A(A)}$ for nonempty A. Every permutation of A fixes its full subset, so naturality makes S_A invariant under all permutations. If it contains one point, transpositions show it contains every point; otherwise

it is empty. Under the map $A \to 1$, naturality says the direct image of S_A equals $\eta_1(1)$, so all nonempty A choose the same case. For an arbitrary subset $B \subset X$, inclusion $i : B \to X$ gives $\eta_{X(B)} = i\big(\eta_{B(B)}\big)$. Therefore either every subset is sent to itself, or every subset is sent to the empty set. Both operations commute with direct images, so these two transformations are exactly the answer.

A tempting dead end. Testing only bijections permits choices depending on cardinality. Naturality under all functions removes those extra choices.

Hypotheses. The power-set functor is covariant by direct image. The proof would be a different problem for inverse image on the opposite category.

Check your argument. Include empty subsets, nonempty sets, arbitrary inclusions, and verification of both resulting transformations.

4 Unit 4 · Universal properties

MM7030-U4.01 · E · 15 min
Learning objective: U4.1 Initial and terminal
Identify initial and terminal objects in sets and in groups.
In sets, the empty set is initial and any singleton is terminal. In groups, the trivial group is initial because its identity must map to the identity, and terminal because there is one homomorphism sending every element to the identity.
Check your argument. Explain uniqueness, not only existence.

MM7030-U4.02 · E · 15 min
Learning objective: U4.1 Products
Describe the product of a two-element set and a three-element set with its projections.
The Cartesian product has six pairs. The maps $(a, b) \mapsto a$ and $(a, b) \mapsto b$ are its projections. For maps $f : X \to A, g : X \to B$, the unique mediator is $x \mapsto (f(x), g(x))$.
Check your argument. Include the universal property.

MM7030-U4.03 · E · 15 min
Learning objective: U4.2 Coproducts
Why is ordinary union not always a coproduct of two sets?
If $A = B = \{*\}$, ordinary union is a singleton. Two maps from A, B into a two-element target can send their elements to different values, and no map from the singleton union can extend both. A tagged disjoint union has two elements and does satisfy the coproduct property.
Check your argument. Use incompatible maps to demonstrate failure.

MM7030-U4.04 · E · 15 min
Learning objective: U4.1 Poset products
Find the product and coproduct of twelve and eighteen in positive integers ordered by divisibility.
The product is six and the coproduct is thirty-six. The universal properties are exactly: common divisors divide the gcd, and the lcm divides every common multiple.
Check your argument. Translate the arrow direction into divisibility.

MM7030-U4.05 · C · 30 min
Learning objective: U4.1 Product uniqueness
Prove two products of fixed objects are uniquely isomorphic compatibly with their projections.
The projections of each product provide a competing pair of maps for the other, yielding unique comparison maps. Each composite preserves both original projections, as does the identity. The uniqueness clause forces these composites to be identities. The comparison map is unique because its projections were prescribed.
Check your argument. Compatibility with projections is essential to the uniqueness claim.

MM7030-U4.06 · C · 30 min
Learning objective: U4.2 Finite biproducts
Prove $V \oplus W$ is both a product and a coproduct in vector spaces.
For a product, send x to $(f(x), g(x))$; linearity and uniqueness follow coordinatewise. For a coproduct, send (v, w) to $a(v) + b(w)$; this is linear and extends the given maps on the two summands. Every vector is $(v, 0) + (0, w)$, so this extension is unique.
Check your argument. Give both universal properties with their different arrow directions.

MM7030-U4.07 · C · 30 min
Learning objective: U4.2 Free vector spaces
Construct the linear extension of a function from $\{a, b\}$ into a vector space, taking a to u and b to v.
The unique extension sends $r[a] + s[b]$ to $ru + sv$. This formula is linear, has the specified basis values, and is forced by linearity for every input.
Check your argument. Explain uniqueness using the basis expansion.

MM7030-U4.08 · C · 30 min
Learning objective: U4.1 Terminal products

Show $A \times 1$ is isomorphic to A whenever 1 is terminal and the product exists.

Let $p : A \times 1 \to A$ and $s = \langle 1_A, ! \rangle : A \to A \times 1$. Then $ps = 1_A$. Both sp and the identity on the product have first projection p and the unique second projection to one; product uniqueness gives $sp = 1$. Thus they are inverse.

Check your argument. Use terminal uniqueness for the second-coordinate comparison.

MM7030-U4.09 · C · 30 min
Learning objective: U4.2 Groups and coproducts

Explain why $C_2 \times C_2$ with its coordinate inclusions is not a coproduct of two copies of C_2 in groups.

Choose the involutions $(12), (23)$ in S_3. Each defines a homomorphism $C_2 \to S_3$. If a map from $C_2 \times C_2$ extended both, the two commuting coordinate generators would map to commuting elements. But $(12)(23) \neq (23)(12)$. Hence no such extension exists, violating the coproduct property.

Check your argument. Use a target and two valid homomorphisms for which extension fails.

MM7030-U4.10 · A · 50 min
Learning objective: U4.1 Product associativity

Construct the canonical isomorphism $(A \times B) \times C \to A \times (B \times C)$ solely from universal properties.

From the source, take its A, B, C coordinate maps a, b, c. Pair b, c to get a map to $B \times C$, then pair this with a. Construct the reverse map by pairing the first two coordinates. Each composite has the same three coordinate maps as the identity; two successive uses of product uniqueness force equality. Thus the maps are inverse and uniquely compatible with coordinates.

Check your argument. Avoid element notation unless specializing to sets.

MM7030-U4.11 · A · 50 min
Learning objective: U4.2 Infinite sums and products

For countably many copies of K, show the finite-support direct sum is a coproduct but generally not the product under its coordinate projections.

Given linear maps from each copy of K to V, send a finite-support tuple to the finite sum of its coordinate images. This defines the unique linear extension, proving the coproduct property. If the same object were a product, the cone from K with every coordinate identity would yield a map sending one to a tuple whose every coordinate is one. This tuple has infinite support and is absent, a contradiction.

Check your argument. Exhibit the cone whose mediator does not exist.

MM7030-U4.12 · A · 50 min
Learning objective: U4.1 Joint detection by projections

Prove product projections jointly determine every arrow into the product. Deduce the naturality of the canonical swap $A \times B \to B \times A$ in both variables.

Decisive idea. Prove a map equality by showing that both maps solve the same uniquely solvable universal problem.

Finding the approach. Universal properties prove equality without element notation. Test the proposed equality by the projections that characterize the product.

Argument. If two arrows into a product have identical projection composites, both mediate the same pair of arrows, so they are equal. The swap is the pairing of the second and first projections. For maps $f : A \to A', g : B \to B'$, the composites swap$(f \times g)$ and $(g \times f)$swap both have projections $g\pi_B$ and $f\pi_A$. Joint detection makes them equal, proving naturality.

Write $s_{A,B} = \langle \pi_B, \pi_A \rangle$. The first projection of either candidate arrow into $B' \times A'$ is $g\pi_B$, and the second is $f\pi_A$. The product property says there is exactly one arrow with those two composites, so the candidates agree. Existence alone is not enough: uniqueness is the part that detects equality. This argument works even in a category whose objects have no underlying elements.

A tempting dead end. An equation that resembles swapping coordinates in sets is not a proof in an arbitrary category. Likewise, agreement after just one projection generally leaves the other component undetermined.

Hypotheses. Both relevant products must exist and their projections must be fixed. No chosen equality between different constructions of the product is required.

Check your argument. Describe both projected composites with their types.

MM7030-U4.13 · A · 60 min · Optional challenge

Learning objective: U4 Transfer: Joining pointed sets

In the category of pointed sets and basepoint-preserving functions, construct the coproduct of two finite pointed sets of sizes m, n. Prove its universal property, compute its size, and compare it with their product.

Decisive idea. Read the compatibility required by the maps, then impose exactly the relation that makes those maps glue.

Finding the approach. The disjoint union gives separate choices of values, but a pointed target forces agreement at the two distinguished points. Enforce only that necessary identification.

Argument. Take the disjoint union of A and B and identify their basepoints to a single point $*$. The two canonical maps preserve basepoints. Given pointed maps $f : A \to X$ and $g : B \to X$, define the mediator on the first summand by f and on the second by g. It is well defined at the identified pair because both values are the basepoint of X. Every equivalence class has a representative in one of the summands, so these rules also force uniqueness. The coproduct has $m + n - 1$ elements. The product is $A \times B$ with basepoint (a_0, b_0) and the usual projections: a pair of pointed maps uniquely gives the pointed pairing. Its size is mn. For $m = n = 2$, the two sizes are three and four, so these objects cannot be isomorphic.

A tempting dead end. A plain disjoint union has two candidate basepoints and cannot make both inclusions pointed without the identification. A Cartesian product solves an incoming-map problem instead.

Hypotheses. Pointed sets are nonempty, and all maps preserve the chosen point. Universal properties must use the morphisms of this category.

Check your argument. Give the mediator, prove it is well defined and unique, and compare the two directions of universality.

5 Unit 5 · Limits and colimits

MM7030-U5.01 · E · 15 min
Learning objective: U5.1 Equalizers
Find the equalizer in sets of $f, g : \mathbb{Z} \to \mathbb{Z}$, $f(n) = n^2, g(n) = n$.
The equalizer is the inclusion $\{0, 1\} \to \mathbb{Z}$. Any function whose composites with f, g agree has all values in this subset and factors uniquely through its inclusion.
Check your argument. Give the structural map and universal factorization.

MM7030-U5.02 · E · 15 min
Learning objective: U5.1 Pullbacks
Find the pullback of $f : \mathbb{Z} \to \mathbb{Z}$, $f(a) = 2a$, and $g : \mathbb{Z} \to \mathbb{Z}$, $g(b) = 3b$, in sets.
The pullback is the set $\{(3k, 2k) : k \in \mathbb{Z}\}$ with coordinate projections. It can be identified with $\mathbb{Z}$ by $k \mapsto (3k, 2k)$. Compatibility of any pair of functions forces each value to be such a pair, giving the unique mediator.
Check your argument. Distinguish the abstract apex from its two projection maps.

MM7030-U5.03 · E · 15 min
Learning objective: U5.2 Coequalizers
In sets, coequalize the two maps from a singleton to $\{a, b, c\}$ selecting a and b.
The coequalizer is the quotient onto the classes $\{a, b\}$ and $\{c\}$. A map from $\{a, b, c\}$ coequalizes the pair exactly when its values at a, b agree, exactly the condition for a unique map from this quotient.
Check your argument. Describe the factorization condition.

MM7030-U5.04 · E · 15 min
Learning objective: U5.1 Empty diagrams
Explain why a limit of the empty diagram is terminal and a colimit is initial.
A cone over the empty diagram has no structural arrows or compatibility conditions. Its universal property asks for a unique arrow from each object to the chosen apex, which is terminality. Reversing arrows gives initiality for the colimit.
Check your argument. Explain why no extra cone data remain.

MM7030-U5.05 · C · 30 min
Learning objective: U5.1 Equalizers are monic
Prove an equalizer arrow is monic using only its universal property.
For equalizer $e : E \to A$, if $eu = ev$, this common composite equalizes the original pair. Both u, v factor it through e, so uniqueness of the factorization yields $u = v$.
Check your argument. Do not assume the category has elements.

MM7030-U5.06 · C · 30 min
Learning objective: U5.1 Pullbacks of monos
Prove a pullback of a monomorphism is monic.
Let the pullback square have $p : P \to A, q : P \to B$, with $f : A \to C$ monic and $fp = gq$. If $qu = qv$, then $fpu = gqu = gqv = fpv$, so $pu = pv$. Both projected composites now agree, and pullback uniqueness gives $u = v$. Thus q is monic.
Check your argument. Identify which projection is the pullback of the given mono.

MM7030-U5.07 · C · 30 min
Learning objective: U5.2 Linear coequalizers
Prove the coequalizer of linear maps $f, g : V \to W$ is $W/\operatorname{im}(f - g)$.
For $h : W \to X$, the condition $hf = hg$ is equivalent to $h(f - g) = 0$, hence to $\operatorname{im}(f - g) \subseteq \ker h$. This is exactly the condition for a unique linear factor through the quotient map. Thus the quotient has the coequalizer property.
Check your argument. Prove the equivalence of the equations and the kernel condition.

MM7030-U5.08 · C · 30 min
Learning objective: U5.2 Pushout of sets

Compute the pushout of inclusions $\{b\} \to \{a, b\}$ and $\{b\} \to \{b, c\}$ in sets.

The resulting set has three elements represented by a, b, c. The two inclusion maps agree on the common b. Any pair of maps from the original sets agreeing at b determines one function on these three elements and conversely, proving universality.

Check your argument. Begin with distinct tagged copies before imposing the identification.

MM7030-U5.09 · C · 30 min
Learning objective: U5.1 Limits in groups

Show the equalizer of group homomorphisms $f, g : G \to H$ is a subgroup of G. Must it be normal?

If $f(a) = g(a)$ and $f(b) = g(b)$, then $f(ab^{-1}) = f(a)f(b)^{-1} = g(a)g(b)^{-1} = g(ab^{-1})$, proving subgroup closure. It need not be normal: in S_3, the equalizer of conjugation by (12) and the identity is its centralizer $\{e, (12)\}$, which is not normal.

Check your argument. Distinguish a general equalizer from a homomorphism kernel.

MM7030-U5.10 · A · 50 min
Learning objective: U5.1 Finite-limit construction

Construct the pullback of $f : A \to C, g : B \to C$ using a product and an equalizer.

Let $e : P \to A \times B$ equalize $f\pi_A, g\pi_B : A \times B \to C$. The composites $\pi_A e, \pi_B e$ form a compatible square. A compatible pair $u : X \to A, v : X \to B$ gives a unique product map $\langle u, v\rangle$ that equalizes the two maps, hence factors uniquely through e. This is precisely the pullback property.

Check your argument. Use both universal properties in sequence and track uniqueness.

MM7030-U5.11 · A · 50 min
Learning objective: U5.2 Gluing may identify more

Give a coequalizer in sets where the initially required identification pairs are not themselves transitive, and compute the actual quotient.

Take $A = \{0, 1\}$, $B = \{a, b, c\}$, with $f(0) = a, g(0) = b$ and $f(1) = b, g(1) = c$. The listed pairs do not include $a \sim c$, but any coequalizing function has $h(a) = h(b) = h(c)$. Thus all three elements must be identified, giving a singleton quotient.

Check your argument. Explain why transitive closure is forced by equality in every target.

MM7030-U5.12 · A · 50 min
Learning objective: U5.1 Pullback pasting

Prove that two horizontally composable pullback squares have a pullback as their outer rectangle.

Decisive idea. Factor the universal mapping problem through the shared edge and apply uniqueness in the same order.

Finding the approach. Introduce a general test object and solve its compatibility equations one square at a time. The order matters: the right square supplies the intermediate data needed by the left.

Argument. Given maps to the outer lower-left and upper-right corners whose composites to the lower-right agree, compose the first map to the lower-middle corner. The right pullback gives a unique map to the upper-middle corner. Its compatibility with the lower-left map is now exactly the hypothesis for the left pullback, yielding a map to the upper-left corner. Any outer mediator must give the same middle map by right-square uniqueness and then the same left map by left-square uniqueness. Thus the outer square is a pullback.

Name the top row $A \to B \to C$ and the bottom row $D \to E \to F$. Given $x : T \to D$ and $z : T \to C$ with equal composites into F, the right square yields $y : T \to B$ with lower composite equal to the image of x in E and right composite z. The left square then yields $w : T \to A$ with projections x, y. For uniqueness, any competing w' has a B-composite solving the same right-square problem, so it equals y; left-square uniqueness then gives $w' = w$.

A tempting dead end. Proving that the outside rectangle commutes establishes only a cone, not a universal cone. Omitting uniqueness is a common incomplete proof.

Hypotheses. Both inner squares are pullbacks and the two rows compose in the indicated direction. The result does not assert that an arbitrary factor of a pullback rectangle is a pullback.

Check your argument. Specify the order of the two factorizations and both uniqueness steps.

MM7030-U5.13 · A · 60 min · Optional challenge

Learning objective: U5 Transfer: Compatible integer measurements

Fix a positive integer n. Find an abelian group P and maps $p, q : P \to \mathbb{Z}$ with p congruent to $2q$ modulo n, such that every pair $f, g : A \to \mathbb{Z}$ with this congruence factors uniquely through p, q. Give a basis for P.

Decisive idea. Parametrize the compatibility equation and verify that the resulting parameterization respects all test maps.

Finding the approach. Translate the congruence into an exact equation with an extra integer. That parameter should both describe the universal object and give its basis.

Argument. Let $P = \{(x, y) \in \mathbb{Z}^2 : n \mid (x - 2y)\}$, a subgroup under coordinatewise addition, and take coordinate projections. Given f, g, the only possible mediator is $h(a) = (f(a), g(a))$. The congruence ensures it lands in P; the coordinate formulas show it is a homomorphism. Its two projections prescribe every value, proving uniqueness. Each element of P is uniquely $k(n, 0) + y(2, 1)$ because $k = \frac{x-2y}{n}$ is an integer. Thus the two displayed vectors form a basis and $P \cong \mathbb{Z}^2$. Under these coordinates $p(k, y) = nk + 2y$ and $q(k, y) = y$.

A tempting dead end. Choosing $P = \mathbb{Z}^2$ with its standard projections ignores the congruence. Choosing only the pairs $(2y, y)$ imposes equality and excludes legitimate multiples of n.

Hypotheses. The positive integer n ensures the integer parameter is unique. The universal property quantifies over all abelian groups A, not only over individual integer pairs.

Check your argument. Prove closure, both existence and uniqueness of mediation, and independence of the two basis vectors.

6 Unit 6 · Representability and Yoneda

MM7030-U6.01 · E · 15 min
Learning objective: U6.1 Yoneda evaluation
For $\eta : \mathcal{C}(A, -) \Rightarrow F$, which element determines η under Yoneda?
It is $\eta_{A(1_A)} \in FA$. Naturality reconstructs each value by $\eta_{B(f)} = F(f)\big(\eta_{A(1_A)}\big)$.
Check your argument. Include the reconstruction formula.

MM7030-U6.02 · E · 15 min
Learning objective: U6.2 Representing the identity
Show the identity functor on sets is covariantly represented by a singleton.
The bijection $\mathrm{Set}(1, X) \to X$ sends f to $f(*)$. Its inverse sends x to the selecting function. Postcomposing with a function $X \to Y$ corresponds to applying it to the selected element, so the bijection is natural.
Check your argument. Verify naturality, not only cardinality.

MM7030-U6.03 · E · 15 min
Learning objective: U6.1 Reconstructing a transformation
Let $x \in FA$. Define the corresponding transformation from $\mathcal{C}(A, -)$ to F.
Define $\eta_{B(f)} = F(f)(x)$. For $g : B \to C$, $F(g)\eta_{B(f)} = F(gf)(x) = \eta_{C(gf)}$, so the family is natural.
Check your argument. Give its formula and naturality check.

MM7030-U6.04 · E · 15 min
Learning objective: U6.2 Universal elements
For the underlying-set functor on groups represented by $\mathbb{Z}$, identify a universal element.
The universal element is $1 \in \mathbb{Z}$. Every $g \in G$ is the image of one under the unique homomorphism $n \mapsto g^n$. This is exactly the universal-element property.
Check your argument. State the unique homomorphism explicitly.

MM7030-U6.05 · C · 30 min
Learning objective: U6.1 Natural binary operations
Classify natural transformations $X \times X \to X$ in sets.
The functor $X \times X$ is naturally $\mathrm{Set}(2, X)$. Yoneda identifies its transformations to the identity with the two elements of 2. Reconstruction evaluates a map $2 \to X$ at the selected element, giving exactly the two projections.
Check your argument. Explain the representable identification and reconstructed operations.

MM7030-U6.06 · C · 30 min
Learning objective: U6.2 Contravariant representables
Given $f : A \to B$, write the natural transformation $\mathcal{C}(-, A) \Rightarrow \mathcal{C}(-, B)$ and prove naturality.
At X, send $u : X \to A$ to $fu : X \to B$. For $k : Y \to X$, $(fu)k = f(uk)$, so the square with the contravariant hom-functor maps commutes.
Check your argument. Keep the test-arrow direction explicit.

MM7030-U6.07 · C · 30 min
Learning objective: U6.2 Universal element category
Show a universal element of a covariant functor is an initial object in its category of elements.
From (A, u) to (X, x), the category of elements permits precisely arrows $f : A \to X$ with $F(f)(u) = x$. Universality gives exactly one such arrow for every target object, which is the definition of initiality. The same unpacking proves the converse.
Check your argument. State both directions.

MM7030-U6.08 · C · 30 min
Learning objective: U6.1 Empty representing set
Classify natural transformations from the constant singleton functor on sets to the identity functor.
Since $\mathrm{Set}(\varnothing, X)$ is always a singleton, Yoneda identifies the transformations with elements of $\varnothing$, so there are none. Directly, the component at the empty set would have to be a function $1 \to \varnothing$, which does not exist.
Check your argument. Give either full argument and recognize the empty-set obstruction.

MM7030-U6.09 · C · 30 min
Learning objective: U6.2 Uniqueness

Prove that naturally isomorphic contravariant representables have isomorphic representing objects.
If $h_A \cong h_B$, the natural isomorphism and its inverse correspond to arrows $f : A \to B, g : B \to A$. Their composites correspond to identity transformations. Faithfulness of the embedding gives $gf = 1_A$ and $fg = 1_B$, so $A \cong B$.
Check your argument. Explain why the arrows are inverse, not merely that both exist.

MM7030-U6.10 · A · 50 min
Learning objective: U6.1 Natural power operations on groups

Classify all natural transformations from the underlying-set functor on groups to itself.
Yoneda identifies these transformations with the underlying set of $\mathbb{Z}$, so each corresponds to one integer n. For $g \in G$, use the homomorphism $f_g : \mathbb{Z} \to G$ sending one to g. Reconstruction gives $\eta_{G(g)} = f_{g(n)} = g^n$. Every such power operation is natural because homomorphisms preserve integer powers. The component functions need not themselves be homomorphisms.
Check your argument. State the last distinction; the target category of the functor is sets.

MM7030-U6.11 · A · 50 min
Learning objective: U6.2 Nonrepresentability

Show the constant empty-set functor from sets to sets is not covariantly representable.
If it were represented by A, then $\mathrm{Set}(A, A)$ would be in bijection with the empty set. But it contains 1_A, even when A is empty. This contradiction proves nonrepresentability.
Check your argument. Use the identity argument for every possible representing set.

MM7030-U6.12 · A · 50 min
Learning objective: U6.2 Natural n-ary operations

For finite sets m, n, classify natural transformations $X^n \to X^m$ on sets, including the empty cases.
Decisive idea. Evaluate at the generic tuple to recover the entire operation, including all empty-index boundary cases.
Finding the approach. Tuples are functions out of an indexing set. This turns a claim about operations on every set into a claim about one finite test set.
Argument. Yoneda gives $\mathrm{Nat}(\mathrm{Set}(n, -), \mathrm{Set}(m, -)) \cong \mathrm{Set}(m, n)$. A function $s : m \to n$ sends an n-tuple x to the m-tuple $j \mapsto x(s(j))$, so all operations select and repeat coordinates. If m is empty there is one transformation; if n is empty and m is nonempty there are none. If both are empty there is one.
Evaluate a proposed transformation at the identity tuple $\mathrm{id}_n : n \to n$. Its value is a function $s : m \to n$. For any tuple $x : n \to X$, naturality with respect to x forces the transformed tuple to be $x \circ s$. Conversely this formula commutes with every map $X \to Y$, proving existence. When n is empty and m is nonempty, evaluating at $X = \varnothing$ would require a function from a singleton to the empty set, so no transformation can exist.
A tempting dead end. Allowing an arbitrary constant coordinate would require choosing an element of every set naturally, including the empty set. The formula must select an existing input coordinate instead.
Hypotheses. The operations are natural with respect to every function of sets. Restricting the category or excluding the empty set changes which arguments are available.
Check your argument. Explain the reversed direction and all empty-set cases.

MM7030-U6.13 · A · 60 min · Optional challenge
Learning objective: U6 Transfer: Families of subsets from pairs

For a function $f : X \to Y$, define $P(f)(S) = f[S]$ and $P(P(f))(\mathcal{A}) = \{f[S] : S \in \mathcal{A}\}$. Classify all families $\theta_X : X \times X \to P(P(X))$ satisfying $P(P(f))\big(\theta_{X(x,y)}\big) = \theta_{Y(f(x),f(y))}$ for every f. How many such families are there?
Decisive idea. Transport a collection attached to a generic pair through two successive direct-image operations; the extra set layer changes the classification.
Finding the approach. Each pair is the image of one generic pair, but the output now has two levels of set structure. Naturality transports a collection of subsets, including its possible empty member.
Argument. Put $D = \{0, 1\}$ and $\mathcal{A} = \theta_{D(0,1)} \subseteq P(D)$. For each $x, y \in X$, let $u : D \to X$ send zero to x and one to y. Naturality forces $\theta_{X(x,y)} = \{u[S] : S \in \mathcal{A}\}$. Conversely, for any fixed collection $\mathcal{A}$ this

formula is natural: applying f to each subset gives precisely the subsets $(f \circ u)[S]$, which is the formula for the pair $(f(x), f(y))$. Distinct collections give distinct values at $(0, 1)$ in D. There are four subsets of D and thus $2^4 = 16$ collections. For example, the empty collection gives $\varnothing$, the collection containing only the empty subset gives $\{\varnothing\}$, and $\{\{0\}, \{1\}\}$ gives $\{\{x\}, \{y\}\}$. When $x = y$, that last output has one member, as direct image requires. The component at the empty set has empty domain and is unique.

A tempting dead end. Counting only the four subsets of D ignores the extra power set. Treating $\varnothing$ and $\{\varnothing\}$ as the same value also loses a distinct natural family. Defining a separate rule for equal inputs generally fails under functions identifying two points.

Hypotheses. Both power-set operations use direct image, and naturality ranges over every function, including identifications. The inputs are ordered, so the two singleton-subset collections remain distinguishable.

Check your argument. Prove determination and naturality, obtain all sixteen collections, and distinguish empty and repeated-input cases.

7 Unit 7 · Adjunctions

MM7030-U7.01 · E · 15 min
Learning objective: U7.1 Adjunction types

For $F : \mathcal{C} \to \mathcal{D}$ left adjoint to G, write the hom-set bijection and the types of the unit and counit components.

The bijection is $\mathcal{D}(FC, D) \cong \mathcal{C}(C, GD)$. The unit has type $\eta_C : C \to GFC$ and the counit $\varepsilon_D : FGD \to D$.

Check your argument. Check every object belongs to the category of the displayed hom set.

MM7030-U7.02 · E · 15 min
Learning objective: U7.1 Currying

Curry the function $f : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$, $f(b, a) = 2b - a$.

The transpose sends b to the function $a \mapsto 2b - a$. Uncurrying evaluates that function at a, recovering the original formula.

Check your argument. State both the curried type and formula.

MM7030-U7.03 · E · 15 min
Learning objective: U7.2 Evaluation

Describe the counit of the product–exponential adjunction in sets.

The counit is $\varepsilon_{X(f,a)} = f(a)$ from $X^A \times A$ to X. It is the uncurried identity on X^A.

Check your argument. Identify the adjunct correspondence producing it.

MM7030-U7.04 · E · 15 min
Learning objective: U7.2 Free-vector-space unit

Describe the unit of the free-vector-space/forgetful adjunction.

At a set X, the unit is $x \mapsto [x]$ from X to the underlying set of the free vector space K^X. It is a set function, not a purported linear map on X.

Check your argument. Keep the source category clear.

MM7030-U7.05 · C · 30 min
Learning objective: U7.1 Power-set adjunction

Prove direct image along $f : X \to Y$ is left adjoint to inverse image on power-set posets.

We need $f[S] \subseteq T$ exactly when $S \subseteq f^{-1}[T]$. The left inclusion says every $s \in S$ has $f(s) \in T$. This is exactly that every $s \in S$ belongs to $f^{-1}[T]$, the right inclusion. Both operations are monotone, and the equivalence gives the poset adjunction.

Check your argument. Check monotonicity and the two-way inclusion equivalence.

MM7030-U7.06 · C · 30 min
Learning objective: U7.1 Floor and inclusion

Let $i : \mathbb{Z} \to \mathbb{R}$ be inclusion as ordered sets. Show i is left adjoint to floor.

Floor is monotone. Since $\lfloor x \rfloor$ is the greatest integer at most x, an integer n satisfies $i(n) \leq x$ exactly when $n \leq \lfloor x \rfloor$. This is the adjunction criterion for posets.

Check your argument. Identify which map is left adjoint; ceiling gives the other adjunction.

MM7030-U7.07 · C · 30 min
Learning objective: U7.2 Free counit

For a vector space V, describe the counit $FUV \to V$ and verify the triangle involving the unit on UV.

The counit sends $\sum_i a_{i[v_i]}$ to $\sum_i a_i v_i$. The unit on the underlying set sends v to $[v]$. Their composite, after applying the forgetful functor as necessary, sends v to v, so $U(\varepsilon_V)\eta_{UV} = 1_{UV}$.

Check your argument. Distinguish formal sums from actual vector-space sums.

MM7030-U7.08 · C · 30 min
Learning objective: U7.2 Transpose formulas

Given unit and counit, write the transpose of $h : FC \to D$ and the inverse transpose of $k : C \to GD$.

The transpose is $G(h)\eta_C : C \to GD$. The inverse transpose is $\varepsilon_D F(k) : FC \to D$. Their domains match because the unit lands in GFC and $F(k)$ lands in FGD. The triangle identities make the two procedures

inverse.
Check your argument. Check the intermediate objects.

MM7030-U7.09 · C · 30 min
Learning objective: U7.1 Diagonal adjunctions
Show the diagonal functor $\Delta : \mathcal{C} \to \mathcal{C} \times \mathcal{C}$ is left adjoint to the binary-product functor when products exist.
The hom set in the product category is $\mathcal{C}(X, A) \times \mathcal{C}(X, B)$. The product property identifies this naturally with $\mathcal{C}(X, A \times B)$. This is exactly $\Delta \dashv \times$, with the product operation regarded as a functor defined by pairing.
Check your argument. Mention naturality of the pairing correspondence.

MM7030-U7.10 · A · 50 min
Learning objective: U7.2 Triangle converse
Prove the two unit–counit transpose formulas are inverse assuming naturality and the triangle identities.
For h, the round trip is $\varepsilon_D FG(h) F\eta_C = h\varepsilon_{FC} F\eta_C = h$. For k, it is $G\varepsilon_D GF(k)\eta_C = G\varepsilon_D \eta_{GD} k = k$. The middle equalities use counit and unit naturality; the final equalities use the corresponding triangle.
Check your argument. Name the naturality instance and triangle used in each direction.

MM7030-U7.11 · A · 50 min
Learning objective: U7.1 Composite adjunctions
If $F \dashv G$ and $F' \dashv G'$ are composable adjunctions, prove $F'F \dashv \mathbb{G}'$.
For appropriate objects C, E, the natural bijections are $\mathcal{E}(F'FC, E) \cong \mathcal{D}(FC, G'E) \cong \mathcal{C}(C, GG'E)$. Their composite is a bijection natural in both variables, so it is an adjunction. The right adjoints appear in reversed composition order.
Check your argument. Specify the three categories and ensure the functors compose.

MM7030-U7.12 · A · 50 min
Learning objective: U7.2 Closure operators
For an adjunction $F : P \to Q$, $G : Q \to P$ of posets, prove GF is monotone, extensive, and idempotent.
Decisive idea. The unit pushes upward and the counit bounds the second pass downward; together they make a closure operator.
Finding the approach. The two adjunction inequalities give opposite bounds on the second application of GF. Idempotence follows by trapping it between equal endpoints.
Argument. Monotonicity follows by composition. The adjunction applied to $Fp \leq Fp$ gives $p \leq GFp$. Applying this to GFp gives $GFp \leq GFGFp$. The counit gives $FG(Fp) \leq Fp$; applying monotone G gives $GFGFp \leq GFp$. Antisymmetry yields equality, proving idempotence.
Extensivity applies to every object, so it can be applied to GFp as well as to p. The counit at Fp gives the opposite inequality after applying monotone G. Finally antisymmetry, not merely transitivity, turns the two inequalities into equality. In a preorder this argument gives equivalence in both directions, rather than literal equality of objects.
A tempting dead end. Monotonicity and extensivity alone do not imply idempotence: $n \mapsto n + 1$ on the integers has both properties but is not idempotent.
Hypotheses. The functors are monotone maps between posets, and the hom-set adjunction is the stated order equivalence. Antisymmetry is needed for equality.
Check your argument. Use both inequalities and antisymmetry.

MM7030-U7.13 · A · 60 min · Optional challenge
Learning objective: U7 Transfer: A relation produces a closure
Given a relation $R \subset X \times Y$, let $E(A)$ be the set of points of Y related to some member of $A \subset X$. For $B \subset Y$, find the greatest subset $U(B) \subset X$ with $E(U(B)) \subset B$. Prove $A \subset U(E(A))$ and $U(E(U(E(A)))) = U(E(A))$.
Decisive idea. The existential image and universal precondition form an order adjunction, whose unit and counit control the second closure step.
Finding the approach. The universal condition on an individual point constructs the greatest permitted subset. An order equivalence then turns the repeated expression into two simple inequalities.
Argument. Set $U(B) = \{x \in X : \text{for all } y, xRy \Rightarrow y \in B\}$. Then $E(A) \subset B$ exactly when $A \subset U(B)$: both say every neighbor of each point of A belongs to B. Thus $U(B)$ is the greatest allowed subset.

Both maps are monotone. Applying the equivalence to $E(A) \subset E(A)$ gives $A \subset UE(A)$. Applying it to $U(B) \subset U(B)$ gives $EU(B) \subset B$. With $B = E(A)$, monotonicity of U gives $UEUE(A) \subset UE(A)$. Extensivity applied to $UE(A)$ gives the reverse inclusion, proving equality. A point with no neighbors is in every $U(B)$, including $U(\varnothing)$.

A tempting dead end. Using 'there exists a neighbor in B' for U is too weak: another neighbor could lie outside B, violating the required inclusion.

Hypotheses. No symmetry or functionality is assumed for the relation. Empty neighbor sets must be treated by the universal quantifier's vacuous truth.

Check your argument. Prove the order equivalence, both inequalities for idempotence, and the empty-neighbor case.

8 Unit 8 · Preservation and applications

MM7030-U8.01 · E · 15 min
Learning objective: U8.1 Preservation direction
Which of products and coproducts must a right adjoint preserve, when they exist?
A right adjoint preserves products because they are limits. No general coproduct-preservation statement follows from being a right adjoint alone.
Check your argument. State the existence qualification and avoid claiming a converse.

MM7030-U8.02 · E · 15 min
Learning objective: U8.1 Terminal objects
Explain why a right adjoint preserves terminal objects.
A terminal object is the limit of the empty diagram, so its image under a right adjoint is terminal by the limit-preservation theorem. Equivalently its maps from X correspond by adjunction to the unique map from FX into the original terminal object.
Check your argument. Give one complete justification.

MM7030-U8.03 · E · 15 min
Learning objective: U8.2 Abelianization
What is the abelianization of an abelian group?
For abelian A, every $aba^{-1}b^{-1} = e$, so the commutator subgroup is trivial and $A^{\mathrm{ab}} = A/\{e\}$ is canonically isomorphic to A.
Check your argument. Identify the quotient map under the isomorphism.

MM7030-U8.04 · E · 15 min
Learning objective: U8.2 Exponentials
In sets, how many functions are in B^A if $|A| = 3$ and $|B| = 2$? Explain its role in currying.
There are $2^3 = 8$ functions. The natural bijection $\mathrm{Set}(X \times A, B) \cong \mathrm{Set}\big(X, B^A\big)$ sends a function of two arguments to its family of functions on A.
Check your argument. Include the representing property, not only the count.

MM7030-U8.05 · C · 30 min
Learning objective: U8.1 Representable preservation
Show $\mathcal{C}(A, -)$ preserves a binary product.
The comparison map sends $h : A \to B \times C$ to $(\pi_B h, \pi_C h)$. Product universality supplies its inverse, pairing the two arrows. Thus it is a bijection respecting projections, which proves preservation of the product in sets.
Check your argument. Identify the actual comparison map.

MM7030-U8.06 · C · 30 min
Learning objective: U8.1 Nonpreservation
Show the free-vector-space functor does not preserve the terminal set.
The terminal set is a singleton; its free vector space is one-dimensional. The terminal vector space is zero-dimensional, so the image is not terminal. This does not contradict left-adjoint preservation, which concerns colimits rather than limits.
Check your argument. Explain the direction of the theorem.

MM7030-U8.07 · C · 30 min
Learning objective: U8.2 Abelianization map
Compute the abelianization of S_3.
Sign kills commutators, so $[S_3, S_3] \subseteq A_3$. The commutator of (12) and (23) is a nontrivial three-cycle, which generates A_3. Thus equality holds and $S_3^{\mathrm{ab}} \cong S_3/A_3 \cong C_2$.
Check your argument. Prove both containments for the commutator subgroup.

MM7030-U8.08 · C · 30 min
Learning objective: U8.2 Free coproducts

Construct the canonical isomorphism $F(A \coprod B) \cong F(A) \oplus F(B)$ for free vector spaces.
Map a basis vector from the A tag to $([a], 0)$ and from the B tag to $(0, [b])$, then extend linearly. The inverse sends each summand basis vector to its tagged basis vector. Both composites are identity on every basis vector, hence on the whole space.
Check your argument. Explain why equality on generators suffices.

MM7030-U8.09 · C · 30 min
Learning objective: U8.2 Implication in a poset

For subsets of S, prove $X \cap A \subseteq B$ iff $X \subseteq (S \setminus A) \cup B$.
If the first inclusion holds and $x \in X$, then either $x \notin A$ or $x \in X \cap A \subseteq B$, proving the right inclusion. Conversely if $x \in X \cap A$, the right inclusion places it in the displayed union, and membership in A excludes its complement, so $x \in B$.
Check your argument. Prove both directions by element membership.

MM7030-U8.10 · A · 50 min
Learning objective: U8.1 Right-adjoint preservation

Prove directly through adjunct transposes that a right adjoint preserves an equalizer.
Let $e : E \to A$ equalize $f, g : A \to B$, and let $u : X \to GA$ satisfy $Gfu = Ggu$. Its transpose $v : FX \to A$ satisfies $fv = gv$ by naturality. There is a unique $w : FX \to E$ with $ew = v$. Transpose w to $z : X \to GE$; naturality gives $Gez = u$. Any second factor would transpose to a second factor through e, impossible. Thus Ge is the equalizer.
Check your argument. Include the uniqueness argument after transposing back.

MM7030-U8.11 · A · 50 min
Learning objective: U8.2 Abelianization adjunction

Prove abelianization is left adjoint to inclusion of abelian groups, including naturality.
A homomorphism $f : G \to A$ with A abelian kills each commutator and hence factors uniquely through G^{ab}. Conversely composing a map from the quotient with its projection gives such an f, and the constructions are inverse. Precomposition by a group homomorphism respects commutators, while postcomposition by an abelian-group homomorphism respects factorization. In each case both routes give the unique factor of the same composite, proving naturality.
Check your argument. Establish the bijection and both variables' compatibility.

MM7030-U8.12 · A · 50 min
Learning objective: U8.1 Equivalences preserve limits

Prove an equivalence of categories preserves all existing small limits.
Decisive idea. Transfer every target cone into the source, solve it there, and transfer uniqueness back along the same isomorphism.
Finding the approach. The target limit must accept cones from objects that are not literally images. Essential surjectivity first replaces such an apex by an isomorphic image object.
Argument. Let F be full, faithful, and essentially surjective, and let L be a limit. A cone in the target with apex X can be transferred along an isomorphism $FA \cong X$. Fullness lifts its legs to arrows from A; faithfulness shows they satisfy the cone equations. Limit universality gives a unique arrow $A \to L$. Applying F and composing with the inverse chosen isomorphism gives a mediator from X. Fullness and faithfulness translate any rival mediator back to one in the source, proving uniqueness.
Choose $i : FA \to X$. Compose target cone legs with i, lift them using fullness, and use faithfulness to reflect every diagram equation. If $u : A \to L$ is the source mediator, the target mediator is $F(u) \circ i^{-1}$. Given a rival $v : X \to FL$, fullness lifts $v \circ i$; faithfulness shows its source composites equal the cone legs, so source uniqueness forces it to equal u. This also shows the resulting mediator is independent of the auxiliary choice of i.
A tempting dead end. A fully faithful functor need not preserve limits tested against objects outside its essential image. Essential surjectivity is what allows every target apex to enter the proof.
Hypotheses. The source diagram has a small indexing category and its limit exists. The argument proves preservation of existing limits, not creation of limits that were absent.
Check your argument. Explain the roles of all three equivalence properties.

MM7030-U8.13 · A · 60 min · Optional challenge

Learning objective: U8 Transfer: A quotient that loses a kernel

On abelian groups define $Q(A) = A/2A$, with induced maps. Prove that Q preserves coproducts and coequalizers, but does not preserve all kernels.

Decisive idea. Use a universal quotient to understand colimits, then test a limit against a map whose injectivity the quotient destroys.

Finding the approach. The quotient has a universal property relative to exponent-two groups. This gives the colimit claims efficiently; an injective map that becomes zero supplies a contrasting kernel example.

Argument. Let B satisfy $2B = 0$. Every homomorphism $A \to B$ kills $2A$ and factors uniquely through $A/2A$. Thus the quotient functor to the full subcategory of exponent-two groups is left adjoint to inclusion. To see preservation in all abelian groups explicitly, $Q(\bigoplus_i A_i) \cong \bigoplus_i Q(A_i)$ by coordinate classes, since elements have finite support. The coequalizer of $f, g : A \to B$ is $B/\operatorname{im}(f-g)$. Applying Q gives $B/(\operatorname{im}(f-g) + 2B)$, also the coequalizer of the induced maps $Q(f), Q(g)$. But multiplication by two on $\mathbb{Z}$ has zero kernel, while its image under Q is the zero endomorphism of $\mathbb{Z}/2\mathbb{Z}$, with kernel the entire nonzero group. Therefore the canonical comparison from the image of the original kernel is not an isomorphism.

A tempting dead end. Being a left adjoint to a subcategory inclusion does not by itself assert every colimit comparison in a larger ambient category. The explicit quotient computations justify the claims in abelian groups.

Hypotheses. The category is abelian groups, so images and quotients compute coequalizers. The counterexample requires preserving the kernel diagram, not merely sending the zero object to zero.

Check your argument. Give the natural quotient formulas for both colimits and compare the two kernels explicitly.

9 Mock assessment 1

Allow 180 minutes. Attempt all six problems and explain every inference. Use the rubric in the handbook for self-assessment; this is original practice, not an official examination.

MM7030-M1.01 · E · 15 min
Learning objective: U1 Cancellation
State the definition of epimorphism and prove every surjective set function is epic.
An arrow $e : A \to B$ is epic if $ue = ve$ implies $u = v$ for all parallel maps out of B. If e is a surjective function, choose for each b any a with $e(a) = b$. Then $u(b) = u(e(a)) = v(e(a)) = v(b)$. Thus $u = v$.
Check your argument. No simultaneous choice of preimages is needed; argue for an arbitrary b.

MM7030-M1.02 · E · 15 min
Learning objective: U4 Universal properties
State the coproduct property and identify the coproduct of two finite sets of sizes three and four.
The coproduct is a tagged disjoint union of size seven. For maps from the two sets to any X, define the mediating function on each tagged component by the given map. It is unique because every element belongs to exactly one component.
Check your argument. Include the maps and uniqueness clause.

MM7030-M1.03 · C · 30 min
Learning objective: U3 Naturality
Prove the swap maps $X \times X \to X \times X$, $(x, y) \mapsto (y, x)$ form a natural isomorphism.
For any $f : X \to Y$, both $(f \times f)\mathrm{swap}_X$ and $\mathrm{swap}_{Y(f\times f)}$ send (x, y) to $(f(y), f(x))$, proving naturality. Each swap composed with itself is the identity, so each component is invertible; the inverse family is the same natural transformation. Thus it is a natural isomorphism.
Check your argument. Check naturality and invertibility independently.

MM7030-M1.04 · C · 30 min
Learning objective: U5 Limits
Construct the pullback in sets of two maps to a singleton and prove its universal property.
For $A \to 1$ and $B \to 1$, every pair (a, b) is compatible, so the pullback is $A \times B$. Given maps from X to A, B, the unique mediator is their pairing. Both composites to the singleton agree automatically, so this is exactly the pullback property.
Check your argument. Explain why there is no remaining compatibility restriction.

MM7030-M1.05 · A · 40 min
Learning objective: U6 Yoneda
Classify natural transformations $X^3 \to X^2$ in sets and count them.
Yoneda gives $\mathrm{Nat}(\mathrm{Set}(3, -), \mathrm{Set}(2, -)) \cong \mathrm{Set}(2, 3)$. Each of the two output coordinates independently selects one of the three input coordinates. There are $3^2 = 9$ transformations, explicitly $(x_1, x_2, x_3) \mapsto (x_i, x_j)$ for $i, j \in \{1, 2, 3\}$.
Check your argument. Justify exhaustiveness with Yoneda.

MM7030-M1.06 · A · 40 min
Learning objective: U7–U8 Adjunctions
Prove a left adjoint preserves initial objects directly from its hom-set bijection.
If I is initial in $\mathcal{C}$ and $F \dashv G$, then for every D the adjunction gives $\mathcal{D}(FI, D) \cong \mathcal{C}(I, GD)$. The right set has exactly one element, so the left does too. This is precisely initiality of FI.
Check your argument. Use all targets D, not one particular object.

10 Mock assessment 2

Allow 180 minutes. Attempt all six problems and explain every inference. Use the rubric in the handbook for self-assessment; this is original practice, not an official examination.

MM7030-M2.01 · E · 15 min
Learning objective: U2 Functors

Prove every functor preserves isomorphisms.

If $gf = 1$ and $fg = 1$, then functoriality gives $FgFf = F(gf) = 1$ and $FfFg = F(fg) = 1$. Thus Ff is an isomorphism with inverse Fg.

Check your argument. Use identity preservation as well as composition preservation.

MM7030-M2.02 · E · 15 min
Learning objective: U7 Unit and counit

For free vector spaces, describe the unit at a set X and counit at a vector space V.

The unit sends x to its basis vector $[x]$ in the underlying set of FX. The counit sends $\sum_i a_{i[v_i]}$ in FUV to the actual vector sum $\sum_i a_i v_i$ in V.

Check your argument. Give the source and target of each map.

MM7030-M2.03 · C · 30 min
Learning objective: U5 Coequalizers

Compute the coequalizer in vector spaces of $f, g : K \to K^2$ given by $f(t) = (t, 0)$ and $g(t) = (0, t)$.

The image of $f - g$ is the line spanned by $(1, -1)$. The quotient is one-dimensional and is identified with K by $(a, b) \mapsto a + b$, whose kernel is that line and which is surjective. This map coequalizes f, g and has the quotient universal property.

Check your argument. The formula also works in characteristic two.

MM7030-M2.04 · C · 30 min
Learning objective: U6 Representability

Show the functor $G \mapsto G \times G$ from groups to sets is represented by a free group on two generators.

Let F_2 be free on a, b. Sending a homomorphism to $(f(a), f(b))$ is bijective by the free-group property. Postcomposition applies a group homomorphism to both entries, so this bijection is natural. Hence the underlying-pair functor is represented by F_2.

Check your argument. Use the free-group universal property, not the direct product of two copies of $\mathbb{Z}$.

MM7030-M2.05 · A · 40 min
Learning objective: U2–U5 Equivalence

Let F be full, faithful, and essentially surjective. Prove it sends a terminal object to a terminal object.

For any target object D, choose an isomorphism $FA \to D$. Maps $D \to FT$ correspond by precomposition to maps $FA \to FT$, and full faithfulness identifies the latter with maps $A \to T$. Since T is terminal, this set is a singleton. Thus every D has exactly one map to FT, proving terminality.

Check your argument. Explain why essential surjectivity is needed to cover every D.

MM7030-M2.06 · A · 40 min
Learning objective: U7–U8 Logic

In the power-set poset of S, identify a right adjoint to intersection with a fixed A and derive its counit inequality.

The right adjoint is $G(B) = (S \setminus A) \cup B$, since $X \cap A \subseteq B$ iff $X \subseteq G(B)$. Its counit is $G(B) \cap A \subseteq B$, which holds because an element in A cannot be in its complement and must therefore belong to B.

Check your argument. State the adjunction equivalence and derive the correctly oriented inequality.